# Supplementary material for the article: Playing Technique Recognition by Time–Frequency Scattering

Changhong Wang, *Student Member, IEEE,* Emmanouil Benetos, *Senior Member, IEEE,* Vincent Lostanlen, and Elaine Chew

This document provides further details regarding datasets and detection results for our work "Playing technique Recognition by Time–Frequency Scattering" [1].

## A. Additional information of the datasets

Table I provides detailed information about the content of the CBFdataset, including the number of ground truth frames, the ratio of the number of frames for the non-technique part over that of the target technique, the ratio of the number of frames over the dimensionality of the feature, number of samples, total duration of the ground truth for each technique, and the total duration of ground truth frames with a single label.

Fig. 1 shows the comparison between glissando played in isolation and that performed in full-length pieces. The isolated one consists of consecutive note changes with approximately equal duration, while the performed one variate from each other in different ways. Player 1 lengthened the first note for the glissando at position 1, while emphasised the beginning of the glissando at position 2 by increasing the intensity. Player 2 performed the glissando together with flutter-tongue. These variabilities necessitate the data collection of playing techniques in context.

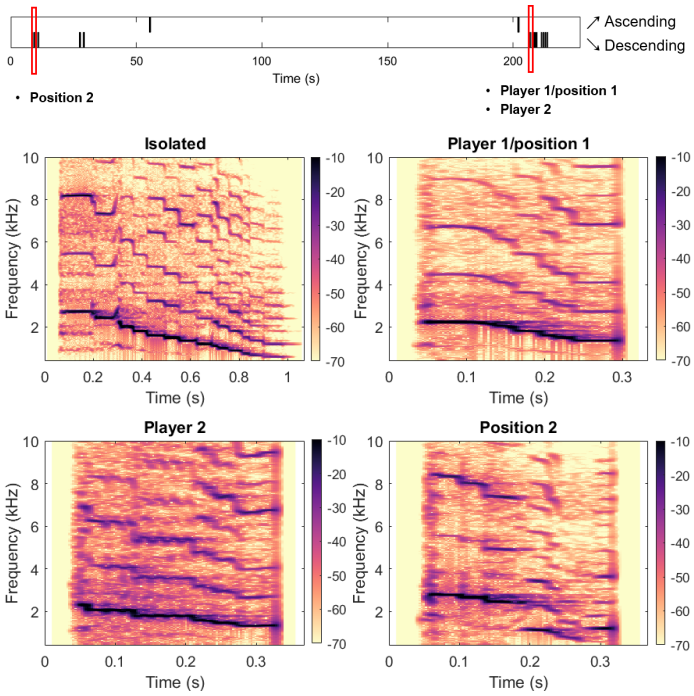


Fig. 1: Variations of glissandi between different players and different temporal positions in the piece *Busy Delivery Harvest*.

To have a better understanding of the detection result in the datasets with playing techniques accurately annotated in real-world pieces, we plot the boxplots of the duration of each type of technique in the CBFdataset and VPset. As can be seen in Fig. 2, PETs exhibit short duration than that of PMTs.

## B. Additional result analysis

Fig. 3 and Fig. 4 are the confusion matrices obtained for the VocalSet and SOL dataset.

## References

[1] C. Wang, E. Benetos, V. Lostanlen, and E. Chew, "Playing technique recogniton by time–frequency scattering," *submitted to IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)*, 2020.

| Techniques | Flutter-tongue | Trills | Vibrato | Tremolo | Acciacatura | Portamento | Glissando |
|---|---|---|---|---|---|---|---|
| No. frames (truth) | 10189 | 4864 | 4449 | 1948 | 2246 | 4096 | 3212 |
| Frame ratio (No:Yes) | 19:1 | 9:1 | 10:1 | 25:1 | 89:1 | 24:1 | 31:1 |
| Frames//Dim (proposed/standard) | 66 / 20 | 12 / 1.4 | 11 / 1.1 | 5 / .5 | 10 / 1.5 | 18 / 2.7 | 15 / 2.3 |
| No. samples | 628 | 197 | 519 | 555 | 441 | 1010 | 490 |
| Duration of ground truth (min) | 7.9 | 15.0 | 13.8 | 6.0 | 1.7 | 6.3 | 5.0 |
| Duration of single-labeled truth (min) | 6.6 | 14.9 | 10.1 | 2.3 | 1.7 | 5.4 | 4.7 |

TABLE I: Frame sizes and dimensionalities of the proposed and the standard scattering features for all seven techniques. Total duration of the CBFdataset is 157 minutes. There are co-articulations with a total duration of 5 minutes in the CBFdataset.

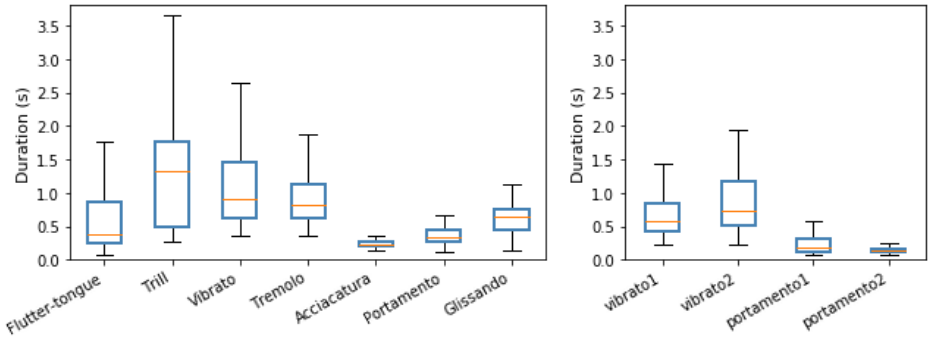


Fig. 2: Boxplots of duration for playing techniques in the CBFdataset and VPset. Top: all seven playing techniques in the CBFdataset; bottom: vibrato and portamento in the VPset (vibrato1=vibrato of erhu and violin; vibrato2=vibrato of instrument excerpts; portamento1 = portamento of erhu and violin; portamento1 = portamento of Beijing opera singing).

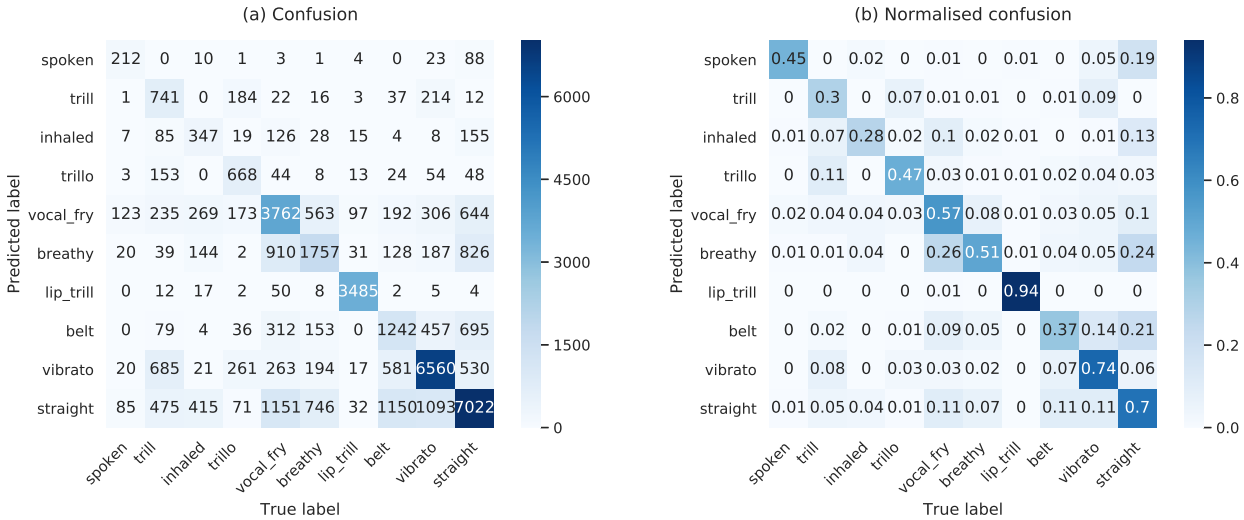


Fig. 3: Confusion matrices obtained for multi-class classification of ten vocal techniques. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

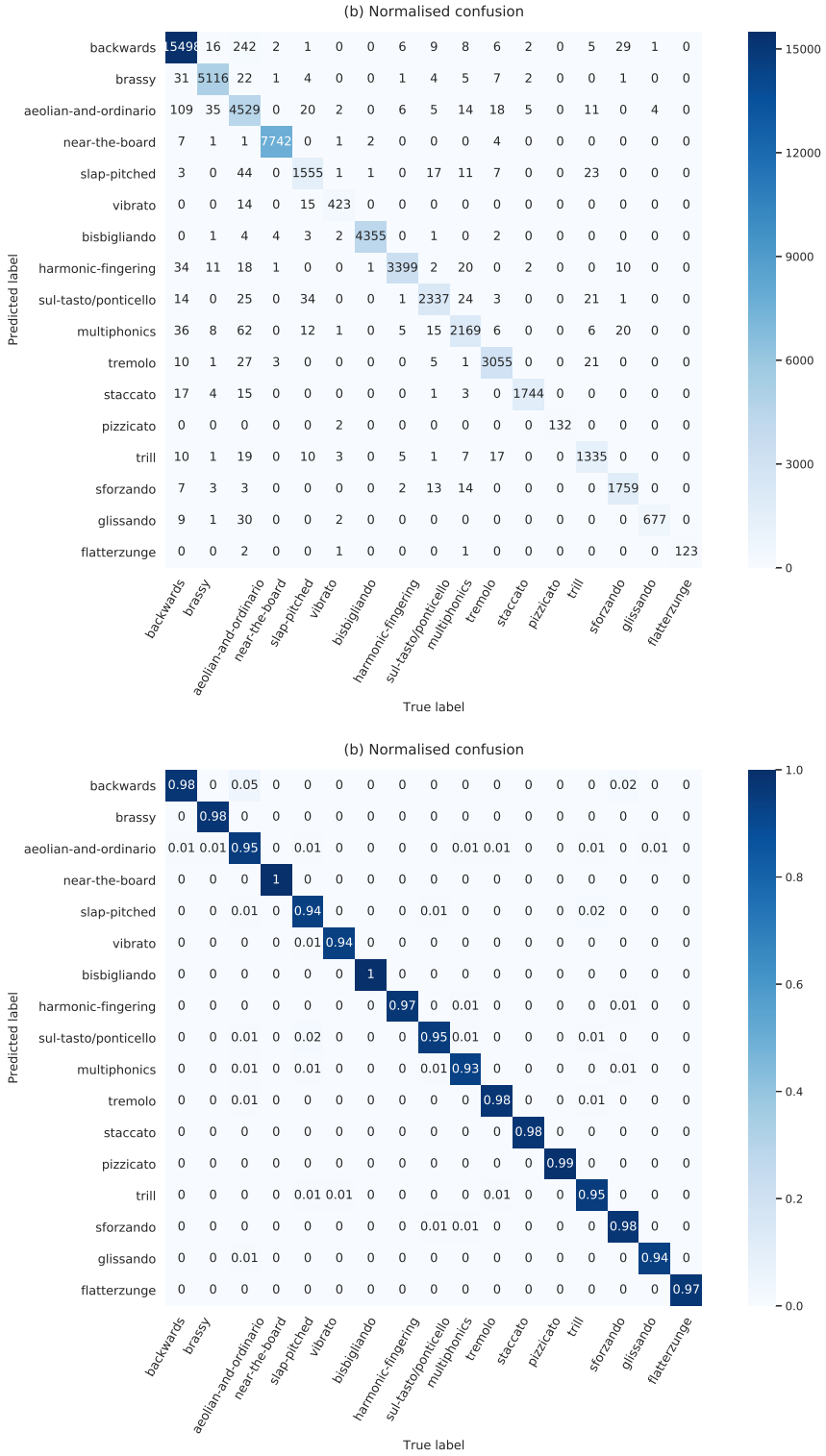


Fig. 4: Confusion matrices obtained for multi-class classification of techniques in the SOL dataset. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.